A2 · Unit 4 · WCH14 Rates, Equilibria and Further Organic Chemistry

Kinetics

At AS you explained qualitatively why concentration, temperature and catalysts change the rate of a reaction. Here the treatment becomes quantitative: rate equations say exactly how the rate depends on each concentration, the orders they contain reveal which species take part in the slowest step of the mechanism, and the Arrhenius equation puts a number on the activation energy.

SPEC 11.1 Key terms in kinetics

WHAT THIS SPEC POINT ASKS

Know the meaning of: rate of reaction, rate equation (rate = k[A]^{m}[B]^{n}, with m and n equal to 0, 1 or 2), order with respect to a substance, overall order, rate constant, half-life, rate-determining step, activation energy, and homogeneous and heterogeneous catalysts.

The rate equation

$$\text{rate} = k[\text{A}]^m[\text{B}]^n$$

$[\text{A}]$: concentration of A / mol dm^{-3} m, n: orders with respect to A and B k: rate constant

The nine definitions

Term	What it means
Rate of reaction	The **change in concentration** of a reactant or product **per unit time**. Units: **mol dm^{-3} s^{-1}**. On a concentration–time graph it is the **gradient** (of a tangent) at that moment.
Rate equation	rate = **k**[A]^{m}[B]^{n}. It links the rate to the concentrations of the species that affect it. It is found **by experiment**, never from the balanced equation.
Order with respect to a substance	The **power** that substance's concentration is raised to in the rate equation (m for A, n for B).
Overall order	The **sum** of the individual orders, m + n.

Term	What it means
Rate constant, k	The constant of proportionality in the rate equation. It is fixed **at a given temperature**: change the temperature (or add a catalyst) and k changes; change a concentration and it doesn't. A bigger k means a faster reaction.
Half-life, $t_{½}$	The **time taken for the concentration of a reactant to fall to half** of its starting value.
Rate-determining step (RDS)	The **slowest step** in a multi-step mechanism. It limits the rate of the whole reaction.
Activation energy, E_a	The **minimum energy** that colliding particles must have for the collision to lead to a reaction.
Homogeneous catalyst	A catalyst in the **same phase** (state) as the reactants, e.g. $H^+(aq)$ in the iodination of propanone (11.6).
Heterogeneous catalyst	A catalyst in a **different phase** from the reactants, usually a solid with gaseous or liquid reactants, e.g. iron in the Haber process (11.11).

A catalyst speeds up a reaction by giving an **alternative route with a lower activation energy**, and it is **regenerated** at the end.

What the orders tell you

Order in A	Rate equation term	[A] × 2	[A] × 3	[A] × 10
0	$[A]^0 = 1$, so A drops out	rate × 1 (no change)	× 1	× 1
1	[A]	rate × 2	× 3	× 10
2	$[A]^2$	rate × 4	× 9	× 100

- A **zero-order** species is not in the rate equation at all, but it still reacts. It just doesn't take part in (or before) the slow step.
- The rate equation can include a **catalyst** (such as H^+) that isn't in the stoichiometric equation. It never includes a species that only appears after the slow step.

Units of k

Rearrange the rate equation, then cancel units: $k = \text{rate} \div ([A]^m[B]^n)$.

Overall order	Units of k
0	mol dm^{-3} s^{-1} (the same as the rate)
1	s^{-1}
2	dm^3 mol^{-1} s^{-1}
3	dm^6 mol^{-2} s^{-1}
4	dm^9 mol^{-3} s^{-1}

Pattern: each extra order adds dm^3 mol^{-1} to the units. Learn the method, not just the table.

Units of k when rate = k[A][B]²

1 Rearrange the rate equation to make k the subject:

$$k = \frac{\text{rate}}{[\text{A}][\text{B}]^2}$$

2 Substitute the units of each quantity:

$$k = \frac{\text{mol}\,\text{dm}^{-3}\,\text{s}^{-1}}{(\text{mol}\,\text{dm}^{-3})(\text{mol}\,\text{dm}^{-3})^2} = \frac{\text{mol}\,\text{dm}^{-3}\,\text{s}^{-1}}{\text{mol}^3\,\text{dm}^{-9}}$$

3 Cancel one mol dm^{-3} from top and bottom, then move what is left on the bottom to the top by changing the signs of its powers:

$$k = \frac{\text{s}^{-1}}{\text{mol}^2\,\text{dm}^{-6}} = \text{dm}^6\,\text{mol}^{-2}\,\text{s}^{-1}$$

The order of the units in the final answer doesn't matter: mol^{-2} dm^6 s^{-1} is equally correct.

WORKED EXAMPLE

Worked example: units of k

rate = k[NO]²[O₂], so k = rate ÷ ([NO]²[O₂]).

Units = mol dm^{-3} s^{-1} ÷ (mol dm^{-3})³ = mol dm^{-3} s^{-1} ÷ (mol^3 dm^{-9})

= mol^{-2} dm^6 s^{-1}, written **dm^6 mol^{-2} s^{-1}** (any order of the units is fine).

MARK-SCHEME WORDING

- Rate equation: "rate = k[NO]²[O₂]". **k must be there**, and concentrations go in **square brackets**. Round brackets are not credited.
- Units of k: "dm⁶ mol⁻² s⁻¹" (third order); "dm³ mol⁻¹ s⁻¹" (second); "s⁻¹" (first). Any order of the units is allowed.
- Effect of temperature on k: "as T increases, k increases"; "a greater proportion of collisions have energy ≥ the activation energy". "More successful collisions" on its own is ignored.
- Homogeneous catalyst: "same state/phase as the reactants"; catalyst: "speeds up the reaction and is regenerated" (WCH14 Jan 2022). If both words are in bold in the question, explain both.

Often asked as: MCQ on units of k or on the effect of changing a concentration; write a rate equation from given orders; calculate k with units; name a catalyst type.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Rate equations with a **fourth overall order**, e.g. rate = k[BrO_3^-][Br^-][H^+]², so k has units dm⁹ mol⁻³ s⁻¹ (WCH14 Jun 2021, Jan 2022).
- **Zero order overall**: rate = k, so k has the same units as the rate (WCH14 Oct 2021).
- The rate equation can contain a species that isn't in the balanced equation as written, e.g. CN^- rather than HCN (WCH14 Oct 2024).

COMMON MISTAKE

Mixing up the **rate constant k** with the **equilibrium constant K**. Le Chatelier and ΔS_{total} = R ln K tell you nothing about k. Explain changes in k with particle energies and E_a (examiner report, Jan 2020).

Leaving k out of the rate equation, or quoting orders from the balanced equation's coefficients.

Answering "explain why the rate increases with temperature" by writing out the Arrhenius equation. Describe what the particles do.

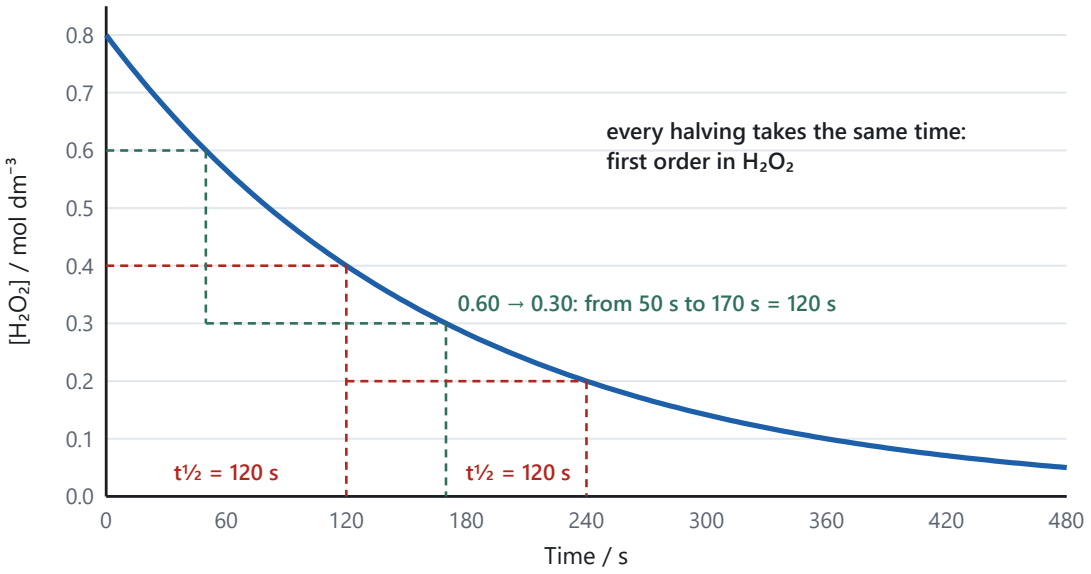
SPEC 11.2 Half-life from a graph: constant half-life means first order

WHAT THIS SPEC POINT ASKS

Find the half-life of a reaction from a suitable graph, and recognise that a constant half-life means the reaction is first order.

Reading half-lives

1. On the concentration–time curve, pick a starting concentration (say 0.80 mol dm^{-3}) and read the time.
2. Find when the concentration has fallen to **half** that value (0.40) and read the time. The difference is one half-life.
3. Repeat from another concentration. It **doesn't have to be consecutive**: 0.60 → 0.30 works just as well as 0.40 → 0.20.
4. Draw the construction lines on the graph. They earn the method mark.



A first-order reaction: 0.80 → 0.40, 0.40 → 0.20 and 0.60 → 0.30 all take 120 s.

Half-life and order

Order	Half-life as the reaction goes on
First	**constant** (it doesn't depend on concentration)
Zero	gets **shorter**
Second	gets **longer**

Only first order has a constant half-life, so "the half-life is constant" is the evidence the mark scheme wants for first order. Radioactive decay is the classic first-order process.

WORKED EXAMPLE

Worked example: half-life to k

From the graph above, t½ = 120 s (three half-lives all 120 s, so first order).

For a first-order reaction, k = ln 2 ÷ t½ = 0.693 ÷ 120 = **5.78 × 10⁻³ s⁻¹**.

Initial rate = k[H_2O_2] = 5.78 × 10⁻³ × 0.80 = **4.6 × 10⁻³ mol dm⁻³ s⁻¹**.

If t½ is read in minutes (say 132 min), convert first: 132 × 60 = 7920 s, so k = 0.693 ÷ 7920 = 8.75 × 10⁻⁵ s⁻¹.

Using a constant half-life

1. A first-order reaction has a half-life of 25 s. The initial concentration of the reactant is 0.64 mol dm^{-3}. What is its concentration after 100 s?

2. The half-life is constant, so count half-lives: 100 s ÷ 25 s = 4 half-lives.

3. Halve the concentration four times:

$$0.64 \rightarrow 0.32 \rightarrow 0.16 \rightarrow 0.080 \rightarrow 0.040 \text{ mol dm}^{-3}$$

Equivalently, $0.64 \times (\frac{1}{2})^4 = 0.64/16 = 0.040 \text{ mol dm}^{-3}$.

MARK-SCHEME WORDING

- "working for at least one half-life shown on graph". Lines on the graph are required.
- "two half-lives … within the range", then "(first order as) the half-life is constant". Stating the value alone does not score the conclusion.
- k = ln 2 ÷ t½ = 0.693 ÷ 120 = 5.78 × 10⁻³ s⁻¹ (IGNORE SF except 1 SF).

Often asked as: show two half-lives on a printed graph and deduce the order; read a half-life (MCQ); turn a half-life into k.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **k = ln 2 ÷ t½** for a first-order reaction (WCH14 Oct 2021 Q17, Oct 2022 Q16).
- The half-life is the time to reach **half the starting concentration**, not half-way to the final concentration. MCQ distractors use the second (WCH14 Jan 2025 Q15).

COMMON MISTAKE

Showing only one half-life, or not writing "constant". Assuming the second half-life must start where the first ended: any halving works (examiner report, Jun 2024).

Leaving k per minute when the question wants s^{-1}.

SPEC 11.3 Choosing a technique to follow the rate

WHAT THIS SPEC POINT ASKS

Pick, and justify, a suitable method to get rate data for a given reaction: titration, colorimetry, mass change, volume of gas, or another suitable technique.

The principle

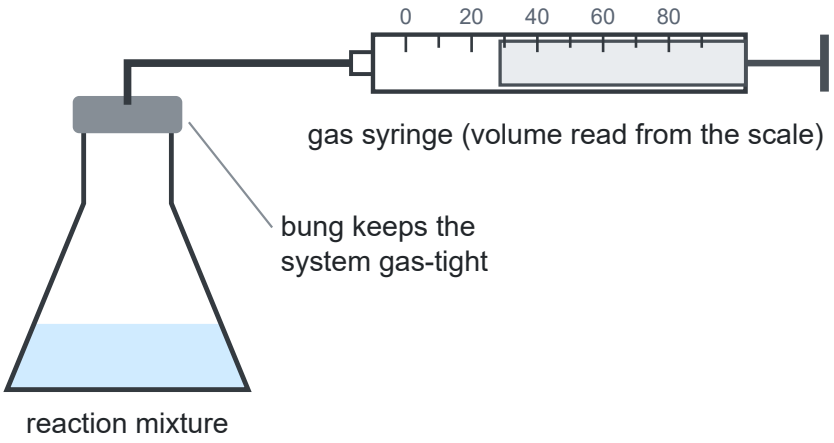
Find a property that **changes as the reaction goes** and is **proportional to the concentration** of one reactant or product. Then measure it against time. The justification must name **what changes in this reaction**.

Technique	Use it when…	How it works	Watch out for
Titration (sample and quench)	an acid, alkali or iodine is used up or made	remove samples at timed intervals, **quench** them (stop the reaction), then titrate	quenching is needed; the reaction itself mustn't make the species you titrate
Colorimetry	a reactant or product is **coloured** (I_2, Br_2, MnO_4^-, NO_2, $Cr_2O_7^{2-}$)	absorbance $\propto$ concentration of the coloured species; use a **calibration curve** and a filter of the **complementary colour**	useless if everything is colourless; a precipitate makes the mixture cloudy
Mass change	a **dense gas** escapes (CO_2 from a carbonate + acid)	flask on a balance; mass falls as gas leaves (loose cotton-wool plug stops spray)	H_2 is too light: the mass change is too small to measure accurately
Volume of gas	any gas is given off	gas syringe (or inverted burette / measuring cylinder over water)	gases that dissolve in water (NH_3, HCl, some CO_2) need a syringe
Other	ions or H^+ change, a precipitate forms	**pH** probe; **electrical conductivity** (number or charge of ions changes); turbidity / "disappearing cross" for a precipitate	conductivity needs a change in total ions

Quenching

To stop a sample reacting before you titrate it:

- **dilute** it with a large volume of ice-cold water (slows it a lot), or
- **remove the catalyst**: add sodium hydrogencarbonate to neutralise an acid catalyst, or filter off a solid catalyst.



Collecting a gas in a gas syringe. Readings of volume against time give a curve whose gradient is the rate of gas production.

WORKED EXAMPLE

Worked example: which method won't work?

$Br_2(aq) + HCOOH(aq) \rightarrow 2H^+(aq) + 2Br^-(aq) + CO_2(g)$

- Colorimetry works: orange-brown bromine is used up.
- Gas volume and mass loss work: CO_2 is given off.
- Conductivity works: ions are produced.
- **Quench and titrate with acid does not**: H^+ is a product, so titrating with acid tells you nothing useful.

MARK-SCHEME WORDING

- Colorimetry: "the reactants are colourless but the product NO_2 is reddish-brown" (WCH14 Oct 2023). Name the coloured species.
- Manganate(VII) titration of H_2O_2: "the concentration (of H_2O_2 is) proportional to the volume of manganate(VII)"; "5:2 ratio" allowed (WCH14 Jun 2025).
- Solid catalyst: quench by "separating the reaction mixture from the catalyst" (WCH14 Jun 2025).

Often asked as: MCQ "which technique is (not) suitable"; justify colorimetry for a named reaction; how to stop a sample reacting.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Colours you need from AS: NO_2 is **brown** (thermal decomposition of Group 2 nitrates), $Br_2(aq)$ orange/yellow, $I_2(aq)$ yellow-brown, MnO_4^- purple (WCH14 Oct 2023 Q15, Jan 2025 Q20).
- Quench an acid-catalysed reaction with **sodium hydrogencarbonate** (WCH14 Oct 2025 Q11).

COMMON MISTAKE

Writing "there must be a colour change" without saying **which** species is coloured. The commonest slip: saying NO is brown and NO_2 colourless (it's the other way round).

Describing a thiosulfate titration when the question is about a clock reaction.

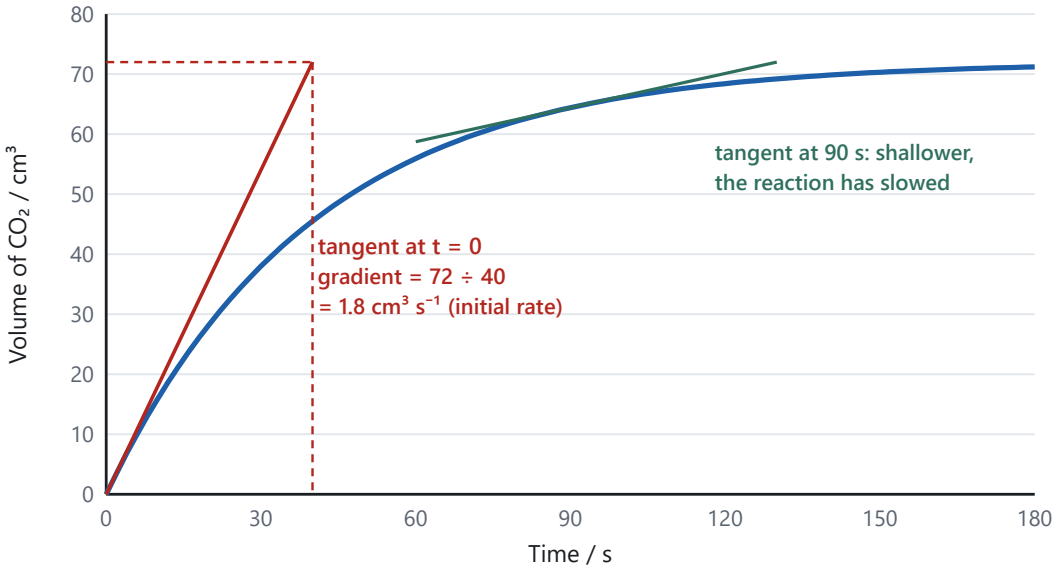
SPEC 11.4 Initial-rate and continuous-monitoring methods

WHAT THIS SPEC POINT ASKS

Understand two ways of getting rate data: (i) the initial-rate method, with separate experiments at different starting concentrations of one reagent (a clock reaction is an acceptable approximation); (ii) continuous monitoring, giving a concentration–time or volume–time graph.

Initial-rate method

- Run **several separate experiments**. In each one change the starting concentration of **one** reagent and keep all the others (and the temperature) the same.
- Measure the rate at the very start, t = 0, when the concentrations are exactly the ones you made up.
- Make up each mixture to the **same total volume** with water, so that changing one volume doesn't dilute the others differently.
- From a concentration–time (or volume–time) curve, the initial rate is the **gradient of the tangent drawn at t = 0**.



The initial rate is the gradient of the tangent at t = 0. Later tangents are shallower because the reactants are being used up.

Clock reactions

A clock reaction gives one time, t: the time for a sudden colour change. That time measures the initial rate because:

- A **small, fixed amount** of a product has to form before the colour appears.
- So little reactant has been used by then that the concentrations have **hardly changed**: the rate is still the initial rate.
- **Rate** $\propto$ **1/t** (shorter time, faster reaction).

In the iodine clock (Core Practical 9b, 11.12): iodine is made slowly; a small amount of **thiosulfate** turns it straight back into iodide. When all the thiosulfate is used up, free iodine meets the **starch**, and the mixture suddenly turns **blue-black**.

Continuous monitoring

- **One** experiment. Measure the concentration (or a property proportional to it) at **many times** as the reaction goes.
- Plot concentration (or volume, or absorbance) against time. Then:
  - read **half-lives** (11.2), or
  - draw **tangents** at several points to get rates at different concentrations, then plot rate against concentration (11.5).
- Examples: gas syringe readings every 10 s; a colorimeter logging absorbance; sampling and titrating (11.6).

WORKED EXAMPLE

Worked example: rate from a clock time

5.0 cm^3 of 0.010 mol dm^{-3} $Na_2S_2O_3$ is in a total volume of 50.0 cm^3. The blue-black colour appears after 40 s. Find the rate of formation of iodine.

Moles of $S_2O_3^{2-}$ = 0.0050 × 0.010 = 5.0×10^{-5} mol

$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$, so moles of I_2 formed before the colour = $5.0 \times 10^{-5} \div 2 = 2.5 \times 10^{-5}$ mol

Concentration of I_2 = $2.5 \times 10^{-5} \div 0.0500 = 5.0 \times 10^{-4}$ mol dm^{-3}

Rate = $5.0 \times 10^{-4} \div 40$ = **1.3×10^{-5} mol dm^{-3} s^{-1}**

MARK-SCHEME WORDING

- Clock chemistry: "sodium thiosulfate reacts with the iodine formed"; "when (all) the thiosulfate is used up the iodine reacts with the starch giving a blue-black colour"; "the reciprocal of the time taken … is a measure of the rate".
- Why it's an initial rate: "the thiosulfate is used up before the concentrations of the other reagents change significantly".
- Colour change: "**colourless to blue-black**" (blue or black allowed). "Clear" is not a colour.
- Initial rate from a graph: "tangent drawn at t = 0" and its gradient, in mol dm^{-3} s^{-1}.

Often asked as: explain how a clock reaction measures an initial rate; state the colour change; calculate a rate from a clock time; draw a tangent at t = 0.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- The iodine clock used in WCH14 is often **hydrogen peroxide + iodide + acid**, not propanone + iodine (WCH14 Jan 2023 Q17, Jan 2024 Q16).
- Moles of I_2 = ½ × moles of thiosulfate; divide by the **mixture volume** and by the **time** (WCH14 Jan 2024 Q16).

COMMON MISTAKE

Dividing moles of iodine by volume but forgetting to divide by time. Using a single point instead of a tangent. Giving the rate per minute when it should be per second.

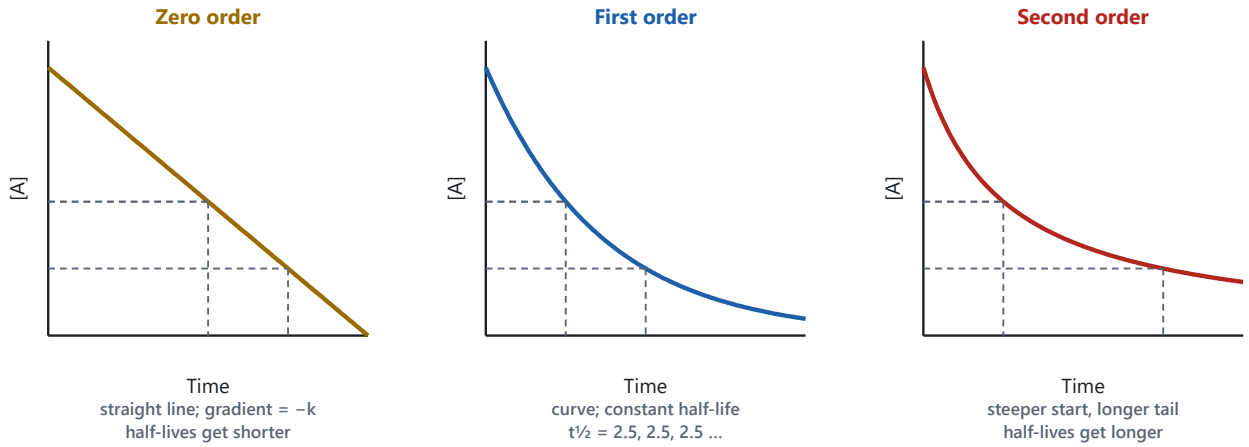
Saying starch reacts with iodide ions. Starch detects **iodine**, I_2.

SPEC 11.5 Deducing orders from graphs and initial-rate data

WHAT THIS SPEC POINT ASKS

Deduce the order (0, 1 or 2) with respect to a substance from (i) a concentration–time graph, (ii) a rate–concentration graph, (iii) initial-rate data.

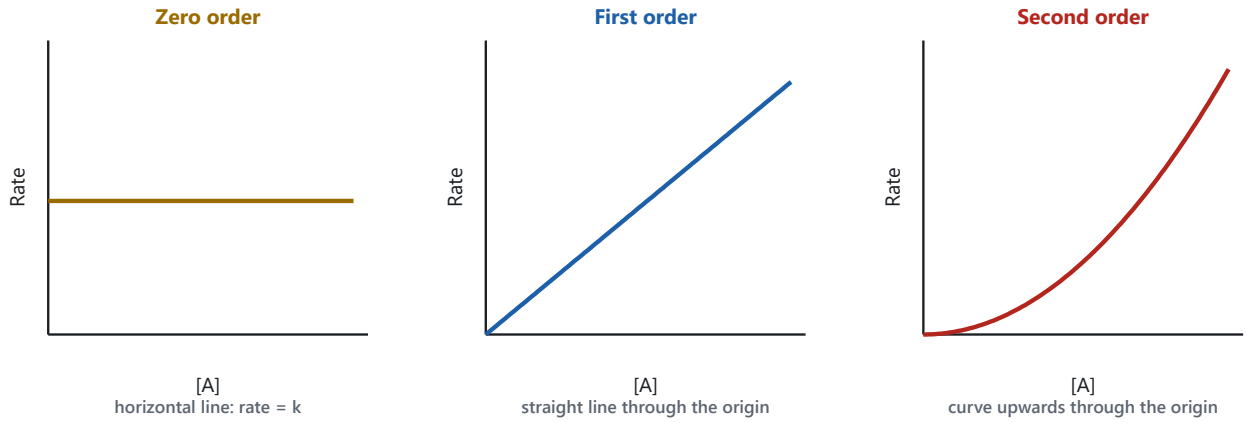
(i) Concentration–time graphs



Zero order: straight line. First order: curve with a constant half-life. Second order: falls fast, then a long tail; half-lives get longer.

- **Zero order**: a **straight line** going down. The gradient (the rate) is constant and equals −k.
- **First order**: a curve with a **constant half-life**.
- **Second order**: a curve that starts steeper and levels off more slowly; **half-lives increase**.

(ii) Rate–concentration graphs



Rate against concentration: horizontal (zero), straight line through the origin (first), upward curve through the origin (second).

- **Zero order**: **horizontal** line. Changing [A] doesn't change the rate.
- **First order**: **straight line through the origin**. Rate ∝ [A]; the gradient is k.
- **Second order**: **upward curve** through the origin. To prove it, plot rate against $[A]^2$: you get a straight line through the origin.
- You get the points by drawing tangents on one concentration–time curve, or from separate initial-rate experiments.

EXAM TIP

Two graphs are straight lines and easy to mix up: **zero order on a concentration–time graph** and **first order on a rate–concentration graph**. Always read the axis labels first.

(iii) Initial-rate tables

1. Find two experiments where **only one** concentration changes.
2. Work out the factor the concentration changes by and the factor the rate changes by.
3. Same factor → first order; factor squared → second order; no change → zero order.
4. Repeat for each reactant. If no pair changes only one concentration, allow for the order you already know (see below).
5. Write the rate equation, then substitute one experiment's values to find k, with units.

WORKED EXAMPLE

Worked example: orders, rate equation and k

$BrO_3^-(aq) + 5Br^-(aq) + 6H^+(aq) \rightarrow 3Br_2(aq) + 3H_2O(l)$

- Exp 1: [BrO_3^-] 0.10, [Br^-] 0.10, [H^+] 0.10; initial rate 1.2×10^{-3} mol dm^{-3} s^{-1}
- Exp 2: [BrO_3^-] 0.20, [Br^-] 0.10, [H^+] 0.10; rate 2.4×10^{-3}
- Exp 3: [BrO_3^-] 0.20, [Br^-] 0.20, [H^+] 0.10; rate 4.8×10^{-3}
- Exp 4: [BrO_3^-] 0.10, [Br^-] 0.10, [H^+] 0.20; rate 4.8×10^{-3}

BrO_3^-: Exp 1 → 2, [BrO_3^-] × 2 with the others constant; rate × 2. **First order.**

Br^-: Exp 2 → 3, [Br^-] × 2 with the others constant; rate × 2. **First order.**

(Exp 1 → 3 would also work: both double and the rate goes × 4. BrO_3^- accounts for × 2, so Br^- must account for the other × 2.)

H^+: Exp 1 → 4, [H^+] × 2; rate × 4 = 2^2. **Second order.**

rate = $k[BrO_3^-][Br^-][H^+]^2$ (fourth order overall)

From Exp 1: $k = 1.2 \times 10^{-3} \div (0.10 \times 0.10 \times 0.10^2)$ = **12 dm^9 mol^{-3} s^{-1}**

Experiment	[NO] / mol dm^{-3}	[H_2] / mol dm^{-3}	Initial rate / mol dm^{-3} s^{-1}
1	5.0×10^{-3}	2.0×10^{-3}	1.25×10^{-5}
2	1.0×10^{-2}	2.0×10^{-3}	5.0×10^{-5}
3	1.0×10^{-2}	4.0×10^{-3}	1.0×10^{-4}

Rate equation for $2NO(g) + 2H_2(g) \rightarrow N_2(g) + 2H_2O(g)$

1. **Order with respect to NO:** compare experiments 1 and 2, where only [NO] changes. [NO] doubles and the rate goes from 1.25×10^{-5} to 5.0×10^{-5}, a factor of $4 = 2^2$.

   So the reaction is second order with respect to NO.

2. **Order with respect to H_2:** compare experiments 2 and 3, where only [H_2] changes. [H_2] doubles and the rate doubles.

   So the reaction is first order with respect to H_2.

3. Write the rate equation. The overall order is 2 + 1 = 3.

$$\text{rate} = k[\text{NO}]^2[\text{H}_2]$$

4. Find k by substituting the values from any one experiment, here experiment 1:

$$k = \frac{\text{rate}}{[\text{NO}]^2[\text{H}_2]} = \frac{1.25 \times 10^{-5}}{(5.0 \times 10^{-3})^2 \times 2.0 \times 10^{-3}} = 250\ \text{dm}^6\,\text{mol}^{-2}\,\text{s}^{-1}$$

Check with another experiment: $250 \times (1.0 \times 10^{-2})^2 \times 4.0 \times 10^{-3} = 1.0 \times 10^{-4}$, *matching experiment 3.*

Experiment	[P] / mol dm^{-3}	[Q] / mol dm^{-3}	Initial rate / mol dm^{-3} s^{-1}
1	0.020	0.030	4.0×10^{-4}
2	0.040	0.030	8.0×10^{-4}
3	0.060	0.090	1.08×10^{-2}

Untangling two changes at once

1. Experiments 1 and 2: only [P] changes. It doubles and the rate doubles, so the reaction is **first order in P**.
2. Experiments 1 and 3: both change. The rate rises by a factor of $1.08 \times 10^{-2} / 4.0 \times 10^{-4} = 27$.
3. [P] has tripled (0.020 → 0.060). Being first order, that alone triples the rate. The rest of the increase must come from Q: $27 \div 3 = 9$.
4. [Q] has also tripled (0.030 → 0.090) and multiplied the rate by 9 = 3^2, so the reaction is **second order in Q**.

$$\text{rate} = k[\text{P}][\text{Q}]^2, \qquad k = \frac{4.0 \times 10^{-4}}{0.020 \times (0.030)^2} = 22\ \text{dm}^6\,\text{mol}^{-2}\,\text{s}^{-1}$$

MARK-SCHEME WORDING

- Name the experiments and say what's held constant: "experiment 2 compared to 1, the concentration of O_2 doubles (while NO remains constant) and the rate doubles … so first order"; "multiplying NO by four … results in an increase in rate of 16 times … second order". Calculations or annotations on the table are accepted as the explanation.

- Graphs you plot: "labelled axes with units", "points cover over half the graph paper", points plotted to ±½ square, then the line.
- First order, rate against concentration: "straight line … beginning at the origin". Concentration–time data: a "smooth curve of best fit", not a ruler.
- Not first order: "graph is not a straight line (through the origin)".

Often asked as: deduce each order from a table with justification; write the rate equation and calculate k; identify or sketch the graph for a given order; plot rate against concentration and conclude.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- A **ten-fold** change in concentration: rate × 10 is first order, × 100 second order (WCH14 Jan 2024 Q16).
- The y-axis of a clock-reaction graph may be 1/t or "rate" rather than concentration (WCH14 Jan 2025 Q18).

COMMON MISTAKE

Quoting orders without justifying them: that loses most of the marks (examiner reports, Jan 2024, Jan 2025).

Drawing a **rate–time** graph when a concentration–time graph is asked for. Joining concentration–time points with straight segments, or ruling a straight line through curved data.

SPEC 11.6 The acid-catalysed iodination of propanone

WHAT THIS SPEC POINT ASKS

Know how to get the data that give the orders with respect to propanone, iodine and hydrogen ions; use the orders to say which species are in the rate-determining step; and suggest a mechanism.

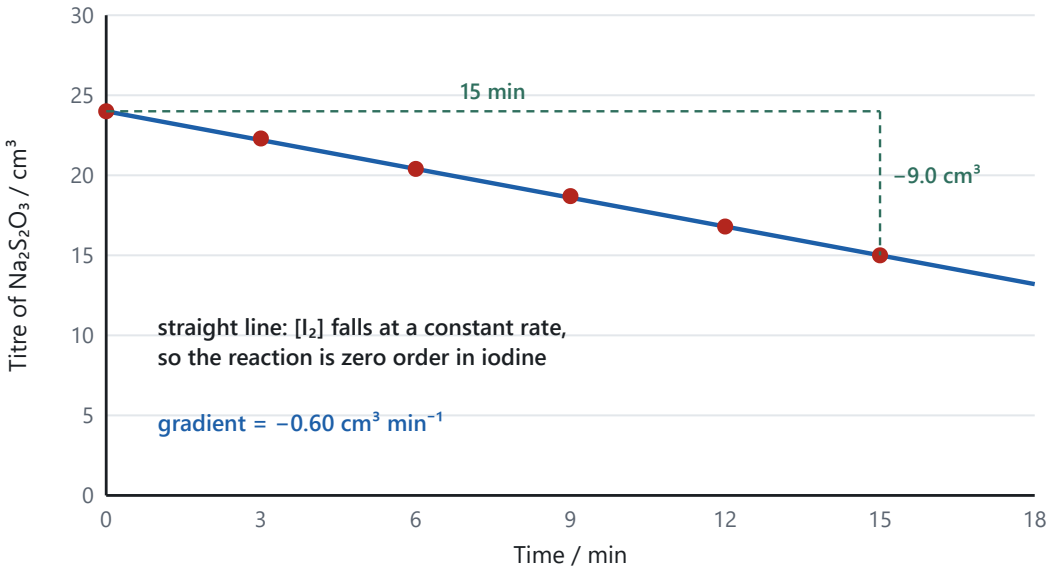
The reaction

$CH_3COCH_3(aq) + I_2(aq) \rightarrow CH_3COCH_2I(aq) + H^+(aq) + I^-(aq)$ (catalysed by H^+)

- The brown iodine colour fades as iodine is used up.
- H^+ is a **catalyst**: it is used in the mechanism and given back. (It is also a product, but with acid in large excess from the start its concentration hardly changes.)

Getting the orders (Core Practical 9a, see 11.12)

- **Iodine**: use **propanone and acid in large excess**, so their concentrations stay almost constant and only $[I_2]$ changes. Follow $[I_2]$ by sampling, quenching with $NaHCO_3$ and titrating with thiosulfate (or with a colorimeter). The plot of $[I_2]$ (or titre) against time is a **straight line**, so the rate doesn't depend on $[I_2]$: **zero order in iodine**.
- **Propanone**: repeat with double the propanone concentration (same acid and iodine). The gradient doubles: **first order**.
- **H^+**: repeat with double the acid concentration. The gradient doubles: **first order**.



Titre against time is a straight line: iodine is used at a constant rate, so the reaction is zero order in I_2.

$$\text{rate} = k[CH_3COCH_3][H^+]$$

What the rate equation says about the mechanism

- **Propanone and H^+** are in the rate equation, so they take part **in or before the rate-determining step**.
- **Iodine** isn't in the rate equation, so it reacts in a **fast step after** the RDS. Changing $[I_2]$ can't speed up a step it isn't part of.
- Overall order 2 → the RDS involves one propanone and one H^+ (directly, or via an intermediate formed from them in a fast step before it).

A mechanism that fits

1. **Fast, reversible**: a lone pair on the carbonyl oxygen accepts H^+.
   $CH_3COCH_3 + H^+ \rightleftharpoons CH_3C(OH^+)CH_3$
2. **Slow (RDS)**: a C–H bond on a CH_3 group breaks; H^+ is lost and the **enol** forms (C=C with an –OH on it).
   $CH_3C(OH^+)CH_3 \rightarrow CH_2{=}C(OH)CH_3 + H^+$

3. **Fast**: the electron-rich C=C of the enol attacks I_2; the O–H loses its H^+ and the C=O re-forms.
   $CH_2=C(OH)CH_3 + I_2 \rightarrow CH_3COCH_2I + H^+ + I^-$

Check it: the steps add up to the overall equation (H^+ cancels out as a catalyst); propanone and H^+ appear up to the slow step; I_2 appears only after it. Step 2 is slow because it breaks a strong C–H bond, so it has a high activation energy.

WORKED EXAMPLE

Worked example: from titre gradient to rate

10.0 cm^3 samples are titrated with 0.0100 mol dm^{-3} $Na_2S_2O_3$. The titre falls by 0.60 cm^3 every minute (graph above).

Thiosulfate per minute = $0.60 \times 10^{-3} \times 0.0100 = 6.0 \times 10^{-6}$ mol

I_2 per minute = $6.0 \times 10^{-6} \div 2 = 3.0 \times 10^{-6}$ mol (in 10.0 cm^3)

Change in $[I_2]$ per minute = $3.0 \times 10^{-6} \div 0.0100 = 3.0 \times 10^{-4}$ mol dm^{-3} min^{-1}

Rate = $3.0 \times 10^{-4} \div 60 =$ **5.0×10^{-6} mol dm^{-3} s^{-1}**

MARK-SCHEME WORDING

- Large excess: "so only changes in iodine concentration affect the rate" / "the concentrations of propanone and acid remain (almost) constant".
- Zero order in iodine: "$[I_2]$ (or volume of thiosulfate) against time gives a straight line", or "rate against $[I_2]$ is a horizontal line".
- Quench: $HCO_3^-(aq) + H^+(aq) \rightarrow CO_2(g) + H_2O(l)$, with state symbols.
- Mechanism: the step with iodine isn't in the rate equation; the steps with propanone and H^+ are. The slow step "involves breaking a (strong) bond / has a high activation energy".

Often asked as: why the excess; which graph shows zero order; quench equation; which steps of a given mechanism fit the rate equation, and which is the RDS.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Why **not** quench with NaOH: base also **catalyses** the reaction (so it wouldn't stop), and in alkali the iodoform reaction / multiple substitution happens (WCH14 Oct 2025 Q11).
- Showing the acid's anion (Cl^-) plays no part: change $[Cl^-]$ with NaCl, or use a different strong acid, and show the rate doesn't change (WCH14 Oct 2025 Q11).
- Mechanism MCQ: protonation of the C=O oxygen is fast; loss of H^+ to form the enol is the slow step (WCH14 Jan 2021 Q7).

COMMON MISTAKE

Concluding that iodine "doesn't react" because it is zero order. It reacts, but after the slow step.

Learning a mechanism by rote and not checking it against the rate equation and the overall equation.

SPEC 11.7 The rate-determining step from a rate equation, and back again

WHAT THIS SPEC POINT ASKS

Deduce the rate-determining step from a rate equation, and the rate equation from a given rate-determining step.

The rules

- The rate equation shows the species (and how many of each) that take part **in the rate-determining step or in fast steps before it**.
- **Order = number of particles of that species** that go into the RDS (directly or via an intermediate made from them before the RDS).
- Species that only react **after** the RDS are **zero order** and not in the rate equation.
- An **intermediate** is never written in the rate equation. Replace it with the species that made it.

Rate equation → slow step

Rate equation	What the slow step contains
rate = k[A]	one A only (A breaks up, e.g. a C–Br bond ionises)
rate = k[A][B]	one A and one B
rate = $k[A]^2$	two A
rate = $k[A]^2[B]$	two A and one B. A three-particle collision is unlikely, so expect $2A \rightleftharpoons A_2$ (fast) then $A_2 + B$ (slow)

Slow step → rate equation

Write the reactants of the slow step. If one is an intermediate, swap it for whatever formed it earlier.

WORKED EXAMPLE

Worked example: is the mechanism consistent?

$2H_2O_2 \rightarrow 2H_2O + O_2$ is found to be **first order** in H_2O_2.

Mechanism X: slow step $H_2O_2 + H_2O_2 \rightarrow \ldots$ (two H_2O_2). This would give rate = $k[H_2O_2]^2$, so it **isn't consistent**.

Mechanism Y: slow step $H_2O_2 \rightarrow H_2O + O$ (one H_2O_2), then fast $O + H_2O_2 \rightarrow H_2O + O_2$. This gives rate = $k[H_2O_2]$ and the steps add up to the overall equation, so it **is consistent**.

Which step is rate-determining?

1. For $2NO(g) + Br_2(g) \rightarrow 2NOBr(g)$ the rate equation is rate = $k[NO]^2[Br_2]$. A two-step mechanism is proposed:

   Step 1: $NO + Br_2 \rightarrow NOBr_2$
   Step 2: $NOBr_2 + NO \rightarrow 2NOBr$

2. Check the steps add up: $NO + Br_2 + NOBr_2 + NO \rightarrow NOBr_2 + 2NOBr$. Cancelling $NOBr_2$ leaves $2NO + Br_2 \rightarrow 2NOBr$, so $NOBr_2$ is an intermediate.

3. If step 1 were the slow step, it would contain one NO and one Br_2, giving rate = $k[NO][Br_2]$. That doesn't match the experiment.

4. If step 2 is slow, it uses one $NOBr_2$ (made from one NO and one Br_2) and one more NO: two NO and one Br_2 in total. That gives rate = $k[NO]^2[Br_2]$, matching the experiment.

   So step 2 is the rate-determining step, and step 1 must be fast.

MARK-SCHEME WORDING

- "the slow step … involves two molecules of H_2O_2 so the mechanism is not consistent with being a first-order reaction" (WCH14 Jun 2025).
- Two checks for "justify": "the steps add up to the overall equation" **and** "the rate equation matches the species in the slow step". Both are needed (WCH14 Oct 2023: only 14% gave both).

Often asked as: MCQ which step fits the rate equation; complete a two-step mechanism; justify whether a proposed mechanism agrees.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- An **intermediate** (e.g. N_2O_2) can appear in the slow step; the rate equation then shows the species that formed it (WCH14 Jan 2026 Q13).

COMMON MISTAKE

Checking only one thing (just the overall equation, or just the rate equation) in a "justify" question.

Giving a stock answer without looking at the steps you've been given.

SPEC 11.8 Deducing a mechanism from the rate equation and the overall equation

WHAT THIS SPEC POINT ASKS

Use the rate equation and the balanced (stoichiometric) equation together to suggest a reaction mechanism.

Method

1. From the rate equation, write the **slow step**: it contains the species (and numbers) in the rate equation.
2. Use **fast steps** before or after it to account for everything else in the overall equation.
3. Check that the steps **add up to the overall equation** (intermediates and catalysts cancel).
4. Check that each step involves **no more than two particles** colliding.

Why most reactions have several steps

A balanced equation like $BrO_3^- + 5Br^- + 6H^+$ has 12 reactant particles. They can't all collide at once: a collision of even **three** particles at the same moment, in the right orientation and with enough energy, is very unlikely. So the reaction must happen in a series of steps involving one or two particles each.

WORKED EXAMPLE

Worked example: three mechanisms

$2NO_2(g) + F_2(g) \rightarrow 2NO_2F(g)$, rate = $k[NO_2][F_2]$

Slow: $NO_2 + F_2 \rightarrow NO_2F + F$

Fast: $NO_2 + F \rightarrow NO_2F$

Sum: $2NO_2 + F_2 \rightarrow 2NO_2F$ ✓ (F is an intermediate)

$NO_2(g) + CO(g) \rightarrow NO(g) + CO_2(g)$, rate = $k[NO_2]^2$ (zero order in CO)

Slow: $NO_2 + NO_2 \rightarrow NO_3 + NO$

Fast: $NO_3 + CO \rightarrow NO_2 + CO_2$

Sum: $NO_2 + CO \rightarrow NO + CO_2$ ✓ (CO only reacts after the slow step)

$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$, rate = $k[NO]^2[O_2]$

Fast, reversible: $2NO \rightleftharpoons N_2O_2$

Slow: $N_2O_2 + O_2 \rightarrow 2NO_2$

Sum ✓. The slow step contains N_2O_2 (made from **2NO**) and O_2, which matches $[NO]^2[O_2]$ without needing a three-particle collision.

WORKED EXAMPLE

Worked example: hydrogen peroxide and iodide

$H_2O_2 + 2I^- + 2H^+ \rightarrow I_2 + 2H_2O$, rate = $k[H_2O_2][I^-]$ (zero order in H^+)

Slow: $H_2O_2 + I^- \rightarrow IO^- + H_2O$

Fast: $IO^- + H^+ \rightarrow HIO$

Fast: $HIO + I^- + H^+ \rightarrow I_2 + H_2O$

Sum: $H_2O_2 + 2I^- + 2H^+ \rightarrow I_2 + 2H_2O$ ✓

MARK-SCHEME WORDING

- Must be multi-step: "there are only 4 particles in the rate equation and 12 in the equation for the reaction", or "collisions with more than two particles are unlikely" (WCH14 Jan 2022).
- "a three-particle collision is unlikely" (hard / difficult / impossible are allowed) (WCH14 Oct 2023).
- 2NO + O_2: "two moles of NO" in step 1 forming N_2O_2, and "O_2 in the rate-determining step": $N_2O_2 + O_2 \rightarrow 2NO_2$. A reversible arrow in step 1 is allowed (WCH14 Jan 2026).

Often asked as: explain why the reaction can't be one step; propose steps that fit a rate equation; say which step is slow.

COMMON MISTAKE

Linking "number of particles" questions to SN1/SN2 when the reaction has nothing to do with halogenoalkanes (examiner report, Jan 2022).

Pasting the stock "five particles can't collide at once" into a question about something else (e.g. the activation energy of a solid reaction).

SPEC 11.9 Rate equations as evidence for SN1 and SN2

WHAT THIS SPEC POINT ASKS

Understand how the rate equations for the hydrolysis of primary and tertiary halogenoalkanes give evidence for the SN2 and SN1 mechanisms.

The evidence

	Primary, e.g. 1-bromobutane	Tertiary, e.g. 2-bromo-2-methylpropane
Rate equation (with OH^-)	rate = k[RBr][OH^-]	rate = k[RBr]
Overall order	2	1
What's in the RDS	the halogenoalkane **and** OH^-	the halogenoalkane **only**
Mechanism	**SN2**: one step	**SN1**: two steps, first one slow
Energy profile	one hump (transition state)	two humps (carbocation intermediate)

S = substitution, N = nucleophilic, 1 or 2 = the number of species in the rate-determining step.

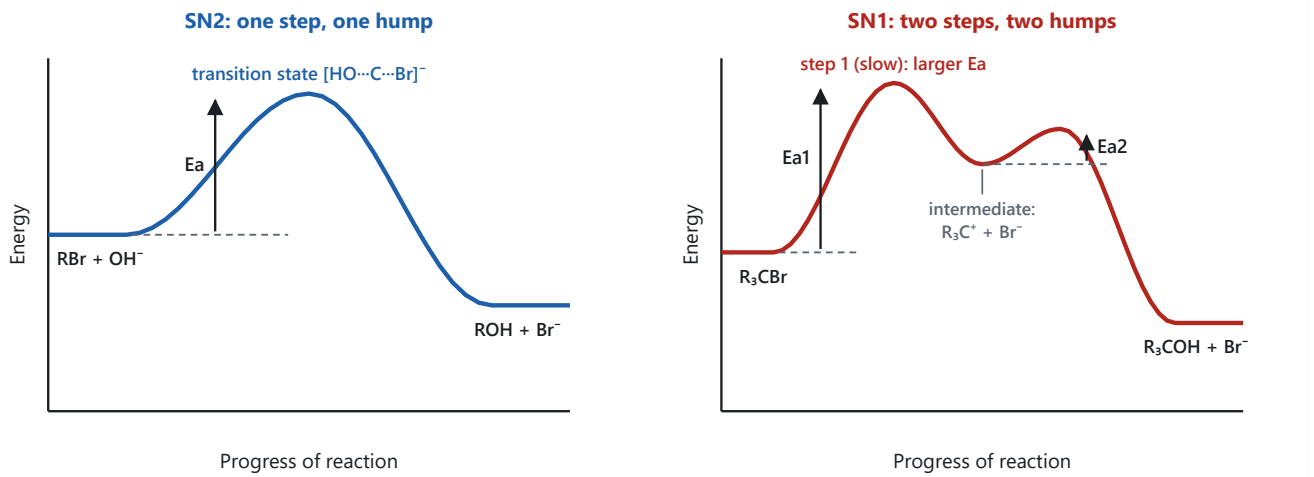
SN2 (primary): one step

- The C–Br bond is polar: C is **δ+**, Br is **δ−**.
- A **lone pair on the oxygen of OH^-** attacks the δ+ carbon, from the side **opposite** the bromine.
- At the same time the **C–Br bond breaks** and both electrons go to Br, which leaves as Br^-.
- Halfway, there is a **transition state** $[HO\cdots C\cdots Br]^-$: five groups round the carbon, two partial bonds, an overall **negative charge** (written outside square brackets).
- Both the halogenoalkane and OH^- are in this one slow step, so the reaction is first order in each.

SN1 (tertiary): two steps

1. **Slow**: the C–Br bond breaks **heterolytically** on its own. Both electrons go to Br; a **tertiary carbocation** $(CH_3)_3C^+$ and Br^- form. Only the halogenoalkane is involved, so rate = k[RBr].
2. **Fast**: a lone pair on OH^- (or on a water molecule) attacks the positive carbon to give the alcohol $(CH_3)_3COH$.

OH^- only reacts in the fast second step, so its concentration doesn't affect the rate.



SN2 has one transition state. SN1 has an intermediate carbocation between two transition states; the first activation energy is the larger, so step 1 is the slow step.

Why primary and tertiary go different ways

- **Tertiary → SN1**: three alkyl groups **push electrons** towards the positive carbon, so the **tertiary carbocation is stabilised**. The three bulky groups also block an OH^- attacking from the back (**steric hindrance**), so SN2 is difficult.
- **Primary → SN2**: a primary carbocation would be very unstable, but the carbon is easy to reach (little steric hindrance).
- **Secondary** halogenoalkanes can react by both.

WORKED EXAMPLE

Worked example: data to mechanism

Hydrolysis of 2-bromobutane with OH^-: doubling [2-bromobutane] doubles the rate; doubling $[OH^-]$ doubles the rate.

rate = $k[C_4H_9Br][OH^-]$, second order overall. Both species are in the slow step, so the mechanism is **SN2**: one step through a transition state.

If instead doubling $[OH^-]$ had **no effect**, rate = $k[C_4H_9Br]$, and the mechanism would be **SN1**: slow ionisation to a carbocation, then fast attack by OH^-.

MARK-SCHEME WORDING

- "rate = k[RX][OH^-]" for primary, "rate = k[RX]" for tertiary; "primary SN2, tertiary SN1".
- Primary: the slow step is the halogenoalkane reacting with OH^- "via a transition state". DO NOT AWARD "intermediate" for transition state.

- Tertiary: the slow step forms "a carbocation (in a step that only involves the halogenoalkane)", then "the hydroxide ion attacks the carbocation in a (fast) second step".
- SN2 diagram: lone pair on O of OH^-; dipole on C–Br; arrow from lone pair to the δ+ carbon; arrow from the C–Br bond to Br; transition state in square brackets with the negative charge outside. A mechanism that doesn't match the given rate equation scores at most 2.
- SN1 profile: "two activation energies imply two steps / an intermediate"; "the first activation energy is larger so the first step is slower".

Often asked as: 6-mark answer comparing a primary and a tertiary bromoalkane (WCH14 Oct 2021 Q18, Jan 2025 Q18); deduce SN1 or SN2 from orders and draw it; MCQ; annotate an energy profile.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Steric hindrance** and **carbocation stability** as the reasons (WCH14 Oct 2025 Q6, Jan 2026 Q3).
- Optical activity: SN2 on one enantiomer gives **inversion** (one optically active product); SN1 goes through a **planar** carbocation that is attacked equally from both sides, giving a **racemic mixture** (WCH14 Oct 2024 Q19, Oct 2025 Q6). See 15.5.

COMMON MISTAKE

Drawing SN2 when the rate equation has no OH^- in it. Drawing SN2 without the transition state, or with arrows that don't start at a lone pair or a bond.

Writing SN1 without the Br^- leaving group, or without saying "slow" then "fast".

Using the "planar carbocation, attacked from above and below" answer in an SN2 question.

SPEC 11.10 Finding the activation energy from experimental data

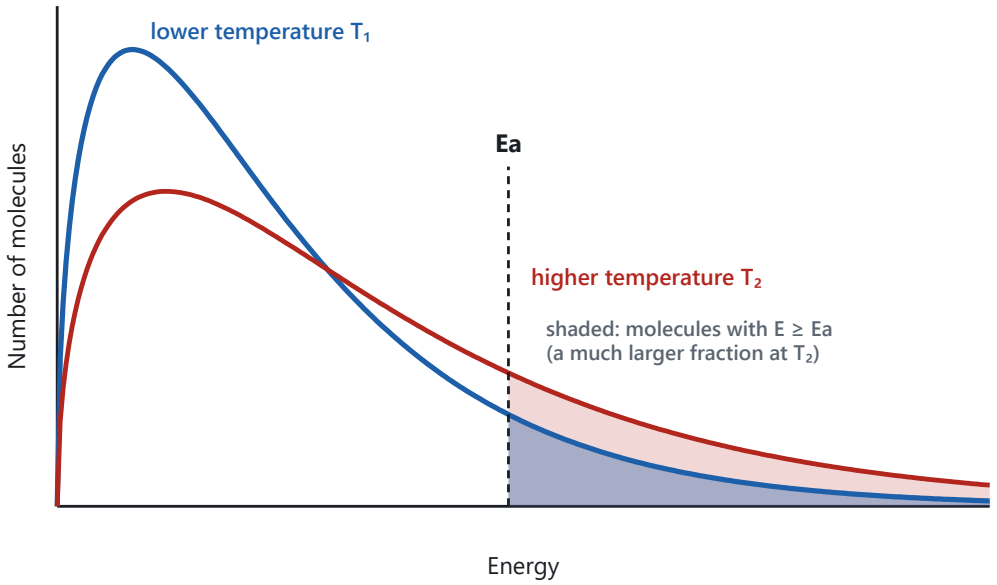
WHAT THIS SPEC POINT ASKS

Use calculations and graphs to find the activation energy of a reaction from experimental data. (The Arrhenius equation is given if you need it.)

Why k increases with temperature

- At a higher temperature the particles have more kinetic energy on average, so they collide more often, but that is a small effect.
- The big effect: a **much larger proportion of collisions have energy ≥ E_a** (the shaded area in the Maxwell–Boltzmann distribution grows a lot).

- So more collisions are successful per second and **k increases**. Roughly, a 10 K rise often doubles the rate.



At the higher temperature the curve flattens and shifts right; the area beyond E_a (molecules able to react) is much larger.

The Arrhenius equation

$$k = A\,e^{-E_a/RT}$$

- k = rate constant; A = the **pre-exponential (Arrhenius) constant**, linked to collision frequency and orientation; E_a in **J mol⁻¹**; R = 8.31 J K⁻¹ mol⁻¹; T in **kelvin**.
- Take natural logs: **ln k = ln A − E_a/(RT)**, which is the straight-line form **ln k = (−E_a/R) × (1/T) + ln A**.

The Arrhenius equation and its logarithmic form

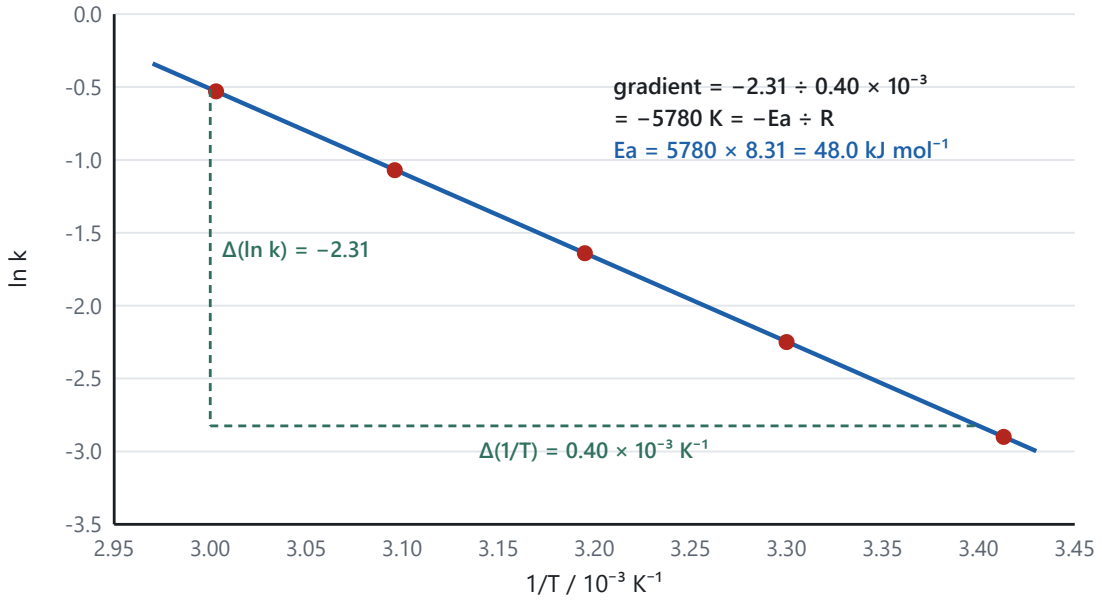
$$k = A\,e^{-E_a/RT} \qquad \ln k = \ln A - \frac{E_a}{R}\cdot\frac{1}{T}$$

k: rate constant A: Arrhenius constant (same units as k) E_a: activation energy / J mol⁻¹
R = 8.31 J K⁻¹ mol⁻¹ T: temperature / K

Graphical method

1. For each temperature, find k (or the rate, or 1/t for a clock reaction: at fixed concentrations all three are proportional to k).
2. Calculate **1/T** (T in K) and **ln k**.

3. Plot **ln k (y-axis) against 1/T (x-axis)**. You get a straight line with a **negative gradient**.
4. **Gradient = $-E_a/R$**, so **E_a = −gradient × R**. The y-intercept is ln A.



Use a large gradient triangle on the best-fit line, not two data points. 1/T values are small, so watch the power of ten.

WORKED EXAMPLE

Worked example: E_a from a graph

- T = 293 K: k = 5.49 × 10^{-2} s^{-1}; 1/T = 3.413 × 10^{-3} K^{-1}; ln k = −2.90
- T = 303 K: k = 0.105 s^{-1}; 1/T = 3.300 × 10^{-3} K^{-1}; ln k = −2.25
- T = 313 K: k = 0.193 s^{-1}; 1/T = 3.195 × 10^{-3} K^{-1}; ln k = −1.64
- T = 323 K: k = 0.342 s^{-1}; 1/T = 3.096 × 10^{-3} K^{-1}; ln k = −1.07
- T = 333 K: k = 0.586 s^{-1}; 1/T = 3.003 × 10^{-3} K^{-1}; ln k = −0.53

Gradient (from the line, between 1/T = 3.00 × 10^{-3} and 3.40 × 10^{-3}) = −2.31 ÷ (0.40 × 10^{-3}) = −5780 K

E_a = −(−5780) × 8.31 = 48 000 J mol^{-1} = **+48.0 kJ mol^{-1}** (3 SF)

WORKED EXAMPLE

Worked example: two temperatures, no graph

The rate constant doubles between 298 K and 308 K. Find E_a.

Write ln k = ln A − E_a/(RT) for each temperature and subtract (ln A cancels):

$\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$

$\ln 2 = (E_a \div 8.31) \times (1/298 - 1/308)$

$0.693 = (E_a \div 8.31) \times 1.0895 \times 10^{-4}$

$E_a = 0.693 \times 8.31 \div (1.0895 \times 10^{-4}) = 52\,900$ J mol^{-1} = **+52.9 kJ mol^{-1}**

WORKED EXAMPLE

Worked example: one point, A given

$k = 3.2 \times 10^{-4}$ s^{-1} at 500 K and $A = 1.0 \times 10^{13}$ s^{-1}. Find E_a.

$\ln k = \ln A - E_a/(RT)$, so $E_a = RT(\ln A - \ln k) = 8.31 \times 500 \times (29.93 - (-8.05))$

$= 8.31 \times 500 \times 37.98 = 157\,800$ J mol^{-1} = **+158 kJ mol^{-1}**

T / K	t / s	1/T / 10^{-3} K^{-1}	ln(1/t)
293	240	3.413	−5.48
303	115	3.300	−4.74
313	58.2	3.195	−4.06
323	30.6	3.096	−3.42
333	16.7	3.003	−2.82

Activation energy from clock-reaction times

1. Calculate 1/T and ln(1/t) for each run (table above) and plot ln(1/t) against 1/T. The points lie on a straight line. Draw the line of best fit.

2. Find the gradient using two points on the line that are far apart, here the first and last:

$$\text{gradient} = \frac{-2.82 - (-5.48)}{(3.003 - 3.413) \times 10^{-3}} = \frac{2.66}{-0.410 \times 10^{-3}} = -6490 \text{ K}$$

3. The gradient equals $-E_a/R$, so:

$$E_a = -\text{gradient} \times R = 6490 \times 8.31 = 53\,900 \text{ J mol}^{-1} = 53.9 \text{ kJ mol}^{-1}$$

The gradient has units of K because ln(1/t) has no units and 1/T is in K^{-1}. Multiplying K by J K^{-1} mol^{-1} gives J mol^{-1}.

Calculating the Arrhenius constant A

1. A first-order reaction has k = 2.5×10^{-4} s^{-1} at 320 K and E_a = 85.0 kJ mol^{-1}. Calculate A.

2. Work out the exponent, with E_a in J mol^{-1}:

$$\frac{E_a}{RT} = \frac{85\,000}{8.31 \times 320} = 31.96 \qquad e^{-31.96} = 1.31 \times 10^{-14}$$

3. Rearrange $k = A\,e^{-E_a/RT}$:

$$A = \frac{k}{e^{-E_a/RT}} = \frac{2.5 \times 10^{-4}}{1.31 \times 10^{-14}} = 1.9 \times 10^{10} \text{ s}^{-1}$$

A has the same units as k, here s^{-1}.

How much faster does a catalyst make a reaction?

1. A catalyst lowers the activation energy of a reaction from 75 kJ mol^{-1} to 50 kJ mol^{-1}. Assuming A is unchanged, by what factor does k increase at 298 K?

2. Divide the two Arrhenius expressions. A cancels:

$$\frac{k_{\text{cat}}}{k} = \frac{e^{-50\,000/RT}}{e^{-75\,000/RT}} = e^{25\,000/(8.31 \times 298)} = e^{10.1} \approx 2.4 \times 10^{4}$$

Lowering E_a by a third makes the reaction about 24 000 times faster, which is why catalysts matter so much in industry.

MARK-SCHEME WORDING

- Graph: ln k on the y-axis, **1/T / K^{-1}** on the x-axis (ln k has no unit), points covering at least half the grid both ways, plotted to ±½ square, straight best-fit line. A capital T, not t.
- Gradient: "measures the increase in ln k and the increase in 1/T"; negative sign; unit **K**.
- "E_a = −gradient × R"; positive answer; units kJ mol^{-1} or J mol^{-1}; usually 3 SF.
- Large E_a → "kinetically stable / rate is low".

Often asked as: plot ln k against 1/T from a table and calculate E_a (most papers); MCQ for the expression (−gradient × R); one-point rearrangement; two-temperature calculation; spot the anomalous point.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Plotting **ln(rate)** or ln(1/t) instead of ln k: same gradient, so the same E_a. Label the axis with what you actually plotted (WCH14 Jan 2024 Q16).
- The **two-temperature** subtraction method (WCH14 Jan 2023 Q17).
- Which bond breaks first in a decomposition: the **weakest** one, from bond enthalpy data (WCH14 Oct 2022 Q16).

COMMON MISTAKE

Axes the wrong way round; a scale that bunches all the points in one corner; no unit on 1/T.

Using one data pair when a graph is asked for. Reading the gradient off two data points instead of the line.

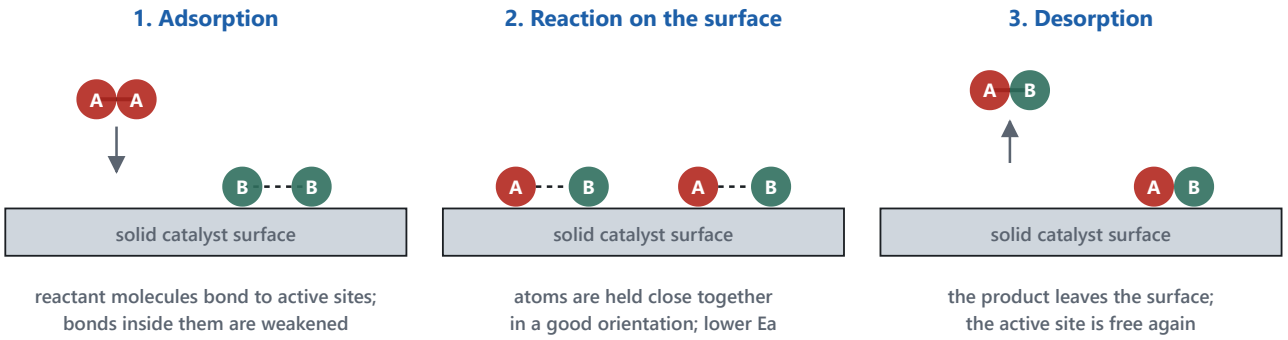
A negative E_a, J and kJ mixed up, more than 3 SF, or °C instead of K.

SPEC 11.11 Solid (heterogeneous) catalysts in industry

WHAT THIS SPEC POINT ASKS

Understand how a solid catalyst works for an industrial gas-phase reaction, by providing a surface for the reaction.

Three steps at the surface



Reactants adsorb, react on the surface, then the product desorbs and frees the active site.

1. **Adsorption**: gas molecules form weak bonds to **active sites** on the catalyst surface. This **weakens the bonds inside the reactant molecules** and holds them close together in a favourable orientation.
2. **Reaction** on the surface, by a route with a **lower activation energy**.
3. **Desorption**: the product molecules break away from the surface, leaving the site free for more reactant.

Making a good industrial catalyst

- **Large surface area**: finely divided powder, pellets, or a thin coat on a **ceramic honeycomb** (catalytic converters). More surface → more active sites → faster reaction, and less of an expensive metal is needed.
- The right **strength of adsorption**: strong enough to weaken bonds, weak enough for products to leave.
- **Poisoning**: impurities (e.g. sulfur compounds; lead in old petrol on catalytic converters) adsorb strongly and block active sites. Feed gases are purified to prevent this.

Process	Catalyst
Haber process: $N_2 + 3H_2 \rightleftharpoons 2NH_3$	iron
Contact process: $2SO_2 + O_2 \rightleftharpoons 2SO_3$	vanadium(V) oxide, V_2O_5
Hydrogenation of C=C (margarine)	nickel
Catalytic converter: CO, NO_x, hydrocarbons	platinum, palladium, rhodium

A catalyst speeds up the forward and reverse reactions **equally**: equilibrium is reached sooner, but the position and K are unchanged (13.4, 13.5).

MARK-SCHEME WORDING

- "reactants adsorb onto the (palladium) catalyst surface"; "this weakens bonds in the reactants"; "products then desorb (from the catalyst surface)" (WCH14 Jun 2021 Q16).

- Catalyst and K: "no effect (on K_c) because the catalyst speeds up both the forward and backward reactions equally" (WCH14 Oct 2025).

Often asked as: explain how a solid catalyst works (3 marks); how to stop a reaction with a solid catalyst (filter it off); catalyst and equilibrium yield.

COMMON MISTAKE

Writing "absorb" for "adsorb". Adsorption is sticking to the **surface**.

Saying a catalyst "increases the yield". It only makes equilibrium arrive faster.

SPEC 11.12 Core Practicals 9a and 9b: iodine–propanone by titration, and an iodine clock

WHAT THIS SPEC POINT ASKS

Core Practical 9a: follow the rate of the iodine–propanone reaction by a titration method. Core Practical 9b: investigate a clock reaction (Harcourt–Esson iodine clock).

CORE PRACTICAL

Core practical 9a: iodine and propanone by titration

Aim: find the order with respect to iodine (and, with extra runs, propanone and H^+).

1. Mix known volumes of **propanone** and **sulfuric (or hydrochloric) acid** in a flask; both are in **large excess** over iodine.
2. Add a known volume of **iodine solution** and start the clock at the moment of mixing. Swirl.
3. Every few minutes, withdraw a **10.0 cm³ sample** with a pipette and run it into a flask of **sodium hydrogencarbonate** solution. This **quenches** the reaction by neutralising the acid catalyst. Note the exact time.
4. Titrate the iodine left in each sample with standard **sodium thiosulfate**. Add **starch** near the end point (when the solution is pale yellow): end point **blue-black → colourless**.
5. Plot titre ($\propto [I_2]$) against time. A **straight line** shows zero order in iodine; the gradient gives the rate (worked example in 11.6).
6. Repeat with a different starting [propanone] (then a different $[H^+]$), everything else the same, and compare gradients to find those orders.

$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$

Control: temperature (water bath), total volume, the volume of each sample, the concentrations of the excess reagents.

Safety: propanone is highly flammable (no flames); iodine solution stains and irritates; wear eye protection.

CORE PRACTICAL

Core practical 9b: an iodine clock

Aim: find the order with respect to one reagent (e.g. iodide) by timing a colour change.

Reaction (one common version): $H_2O_2(aq) + 2I^-(aq) + 2H^+(aq) \rightarrow I_2(aq) + 2H_2O(l)$, slow

Clock step: $I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$, very fast

1. In a beaker, mix potassium iodide, sulfuric acid, a **small fixed volume** of sodium thiosulfate, starch and water.
2. Add the hydrogen peroxide, start the clock and stir.
3. Stop the clock when the mixture suddenly turns **blue-black**. Record t.
4. Repeat with different volumes of KI, making up the **same total volume** with water so only $[I^-]$ changes. Keep the temperature constant.
5. Use **1/t as the rate**. Plot 1/t against $[I^-]$ (or volume of KI): a straight line through the origin shows first order.

Why it works: the thiosulfate removes iodine as fast as it forms, so no blue colour appears. When the thiosulfate runs out, the next iodine formed reacts with starch. The thiosulfate is used up **before the other concentrations have changed much**, so 1/t measures the **initial rate**.

Another version uses peroxodisulfate: $S_2O_8^{2-} + 2I^- \rightarrow 2SO_4^{2-} + I_2$.

Making the results reliable

- Same total volume in every run, measured with burettes or graduated pipettes.
- Thermostat the solutions before mixing; record the temperature.
- Start timing at the moment of mixing; judge the colour change against a white tile. Human reaction time matters most for short times, so choose concentrations that give times of tens of seconds at least.
- Repeat each run and take a mean.

MARK-SCHEME WORDING

- Clock: "sodium thiosulfate reacts with the iodine formed"; "when the thiosulfate is used up the iodine reacts with starch giving a blue-black colour"; "1/time is a measure of rate".
- Large excess: "so the concentrations of propanone and acid remain (almost) constant".
- Colour change in the titration end point and in the clock: give both colours in the right order.

Often asked as: describe how a clock reaction measures an initial rate; calculate rates from clock times; use titration data to show zero order in iodine; explain each step of the method (WCH14 Jan 2023 Q17, Jan 2024 Q16, Oct 2025 Q11).

COMMON MISTAKE

Saying the blue-black colour appears "when the iodine is used up". It appears when the **thiosulfate** is used up.

Changing the volume of one reagent without making up the total volume with water: then every concentration changes.

SPEC 11.13 Core Practical 10: finding the activation energy

WHAT THIS SPEC POINT ASKS

Core Practical 10: find the activation energy of a reaction.

Principle

Keep **every concentration the same** and change only the **temperature**. Then the rate (or 1/t for a clock reaction) is proportional to k, so ln(1/t) against 1/T gives a line of gradient $-E_a/R$.

CORE PRACTICAL

Core practical 10: method in brief

A common version uses a clock reaction between bromide and bromate(V) ions in acid:

$5Br^- + BrO_3^- + 6H^+ \rightarrow 3Br_2 + 3H_2O$

A small fixed amount of **phenol** removes the bromine as it forms. When the phenol is used up, bromine **bleaches methyl red** indicator: time how long the red colour takes to disappear.

1. Put the bromide/bromate mixture, phenol and methyl red in one boiling tube and the sulfuric acid in another.
2. Stand both tubes in a **water bath** until they reach the bath temperature. Record it with a thermometer in the mixture.
3. Mix, start the clock, and stop it when the red colour disappears. Record t.
4. Repeat at about five temperatures between room temperature and about 60 °C, with **identical volumes** each time.
5. Calculate 1/T (T in K) and ln(1/t). Plot ln(1/t) against 1/T; E_a = −gradient × R.

Other versions: the peroxodisulfate/iodide clock at different temperatures; magnesium ribbon in hydrochloric acid (time for the ribbon to disappear).

Safety: phenol is toxic and corrosive (gloves, eye protection); bromine fumes; hot water.

WORKED EXAMPLE

Worked example: from clock times to E_a

At 20 °C, t = 125 s; at 40 °C, t = 34.6 s.

T_1 = 293 K, $1/T_1 = 3.413 \times 10^{-3}$ K^{-1}, $\ln(1/t_1) = \ln(0.00800) = -4.83$

T_2 = 313 K, $1/T_2 = 3.195 \times 10^{-3}$ K^{-1}, $\ln(1/t_2) = \ln(0.0289) = -3.54$

Gradient = $(-3.54 - (-4.83)) \div (3.195 \times 10^{-3} - 3.413 \times 10^{-3}) = 1.29 \div (-0.218 \times 10^{-3}) = -5920$ K

$E_a = 5920 \times 8.31 = 49\,200$ J mol^{-1} ≈ **+49 kJ mol^{-1}**

(In the exam you'd plot all five points and take the gradient from the line.)

Errors and improvements

- Temperature drifts while the reaction runs: use a thermostatted bath; record the temperature at the start and end and use the mean.
- Judging the end point: use a white tile; a colorimeter removes the judgement.
- Mixing time is significant at high temperatures (short times): choose concentrations that give longer times.

MARK-SCHEME WORDING

- The practical procedure is rarely examined; the **data analysis** is (see 11.10): axes, scale, best-fit line, gradient with sign and unit (K), E_a = −gradient × R, sign, units, SF.

Often asked as: plot ln k or ln(rate) against 1/T from given data and find E_a (WCH14 Jan 2024 Q16, Oct 2025 Q2).

COMMON MISTAKE

Changing a concentration as well as the temperature: then the rate isn't proportional to k.

Using °C for T, or forgetting that ln(1/t) is negative when t > 1 s.

Quick checks

QUICK CHECK

The rate equation for a reaction is rate = $k[X]^2[Y]$. What are the units of k?

- $mol^2\ dm^{-6}\ s^{-1}$
- $dm^3\ mol^{-1}\ s^{-1}$
- $dm^6\ mol^{-2}\ s^{-1}$
- $mol\ dm^{-3}\ s^{-1}$

The overall order is 3. k = rate/($[X]^2[Y]$) = (mol dm^{-3} s^{-1})/(mol $dm^{-3})^3$ = mol^{-2} dm^6 s^{-1}. The second option is for a second-order reaction.

QUICK CHECK

A reaction is first order with respect to A and second order with respect to B. Both concentrations are doubled at the same time. By what factor does the rate increase?

- 3
- 4
- 6
- 8

Doubling [A] multiplies the rate by 2; doubling [B] multiplies it by 2^2 = 4. The effects multiply: 2 × 4 = 8. Adding the orders (to get 3) or adding the factors (to get 6) are the usual slips.

QUICK CHECK

A first-order reaction has a half-life of 1400 s. Starting from 1.0 mol dm^{-3}, what is the concentration after 4200 s?

- 0.33 mol dm^{-3}
- 0.125 mol dm^{-3}
- 0.25 mol dm^{-3}
- 0.50 mol dm^{-3}

4200 s is three half-lives, because a first-order half-life is constant: 1.0 → 0.50 → 0.25 → 0.125 mol dm^{-3}. Dividing by 3 instead of halving three times gives the wrong answer of 0.33.

QUICK CHECK

The iodination of propanone has rate = k[CH_3COCH_3][H^+]. Which statement must be true?

- Iodine is a catalyst for the reaction.
- Iodine reacts in a step after the rate-determining step.
- The reaction has a single-step mechanism.
- Doubling [I_2] halves the rate.

Iodine is a reactant, but it is zero order, so it is not involved in the slow step and must react afterwards. A zero-order reactant means there are at least two steps. Changing [I_2] has no effect on the rate.

QUICK CHECK

A graph of ln k against 1/T for a reaction has a gradient of −7500 K. What is the activation energy? (R = 8.31 J K^{-1} mol^{-1})

- [x] 62.3 kJ mol^{-1}
- [ ] −62.3 kJ mol^{-1}
- [ ] 902 kJ mol^{-1}
- [ ] 0.90 kJ mol^{-1}

Gradient = $-E_a/R$, so E_a = 7500 × 8.31 = 62 300 J mol^{-1} = 62.3 kJ mol^{-1}. Activation energy is always positive; dividing by R instead of multiplying gives 0.90.

QUICK CHECK

In a clock reaction, why can 1/t be used as a measure of the initial rate?

- [ ] Because the reaction is zero order, so its rate never changes.
- [ ] Because the thiosulfate is a catalyst that keeps the rate constant.
- [x] Because the same small amount of product forms in every run, before the reactant concentrations have changed much.
- [ ] Because the colour appears when half of the reactants have been used up.

Rate = amount formed ÷ time. The amount (fixed by the thiosulfate) is the same in every run, so the rate is proportional to 1/t. Keeping that amount small means the concentrations are still close to their starting values, so it is an initial rate.

QUICK CHECK

Increasing the pressure of a gas reacting on a solid catalyst has no effect on the rate. What is the best explanation?

- The catalyst has been poisoned.
- The activation energy has risen to compensate.
- Gases do not adsorb onto solid surfaces.
- All the active sites on the catalyst surface are already occupied.

Reaction only happens at active sites. Once they are saturated, extra gas molecules have nowhere to adsorb, so the rate stays the same and the reaction is zero order in that gas.